Part 3

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Iteration 0

* features (X) : $x = ((1,3),(4,10))$

Let $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

* targets : $y = (5,6) \Rightarrow y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

* weights (m) : $m = [-1,2] \Rightarrow m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

* bias (b) : $b = [1,1] \Rightarrow b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

* Number of samples : $N = 2$

* learning rate : $\alpha = 0.01$

$$y = m_1 x_1 + m_2 x_2 + b$$

1. Compute predicted values $\hat{y}$

$$\hat{y} = mX + b$$

* calcule mX

$$mX = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1\times(-1)+(3\times 2) \\ 4(-1)+(10\times 2) \end{bmatrix} = \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

$$\Rightarrow \boxed{mX = \begin{bmatrix} 5 \\ 16 \end{bmatrix}}$$

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$$\hat{y} = mX + b \Rightarrow \hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\boxed{\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}}$$

* calculate error

$$\hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\boxed{\hat{y} - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}}$$

* MSE

$$J = \frac{1}{2}(1^2 + 11^2) = \frac{122}{2} = 61 \Rightarrow \boxed{J = 61}$$

2. Derive the gradient using the chain rule

$$J = \frac{1}{N}\sum(\hat{y} - y)^2$$

Chain rule for m $\left(\frac{\partial J}{\partial m}\right)$

$$\frac{\partial J}{\partial m} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial m}$$

$$\frac{\partial J}{\partial \hat{y}} = \left(\frac{1}{N}(\hat{y} - y)^2\right)'$$

$$\frac{\partial J}{\partial \hat{y}} = 2 \times \frac{1}{N} \times 1(\hat{y} - y)$$

$$\Rightarrow \boxed{\frac{\partial J}{\partial \hat{y}} = \frac{2}{N}(\hat{y} - y)}$$

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(b) ~~$\frac{\partial \hat{y}}{\partial b}$~~ inner derivative

(b) $\frac{\partial \hat{y}}{\partial m} = (mx+b)' = x + 0 = x$

$$\boxed{\frac{\partial \hat{y}}{\partial m} = x}$$

(c) $\frac{\partial J}{\partial m} = \frac{2}{N} \cdot x^T (\hat{y} - y)$

* Chain rule for weights m

$$\frac{\partial J}{\partial m} = \frac{\partial \hat{y}}{\partial m} \cdot \frac{\partial J}{\partial \hat{y}}$$

(a) outer derivative

$$\frac{\partial J}{\partial \hat{y}} = \left(\frac{1}{N}(\hat{y} - y)^2\right)' = 2 \times \frac{1}{N}(\hat{y} - y)$$

$$\boxed{\frac{\partial J}{\partial \hat{y}} = \frac{2}{N}(\hat{y} - y)}$$

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(b) inner derivative

$$\frac{\partial \hat{y}}{\partial m} = 1$$

(c) $$\frac{\partial J}{\partial b} = 1 \times \frac{2}{N}(\hat{y} - y)$$

$$\boxed{\frac{\partial J}{\partial b} = \frac{2}{N}(\hat{y} - y)}$$

3. Calcule numerical gradients

* gradient for m : $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = \frac{2}{2}\begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}\begin{bmatrix} 1 \\ 11 \end{bmatrix} = 1 \times \begin{bmatrix} 1\times1 + (4\times11) \\ (3\times1) + (10\times11) \end{bmatrix} = \begin{bmatrix} 1+44 \\ 3+110 \end{bmatrix}$$

$$\boxed{\frac{\partial J}{\partial m} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}}$$

* gradient for b

$$\frac{\partial J}{\partial b} = \frac{2}{N}(\hat{y} - y) = \frac{2}{2}\begin{bmatrix} 1 \\ 11 \end{bmatrix} = 1 \times \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\boxed{\frac{\partial J}{\partial b} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}}$$

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4. Update parameters m and b

New = old - (α x gradient)

* Update m

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.45 \\ 1.13 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$\boxed{m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}}$$

* Update b

$$b = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.01 \\ 0.11 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$\boxed{b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}}$$

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1st iteration

* features $x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

* targets $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

* weights $m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$

* bias $b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$

① Compute predicted values $\hat{y}$

$$\hat{y} = mx + b$$

* calculate mx

$$mx = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.85 \end{bmatrix} = \begin{bmatrix} 1\times(-1.45) + (3\times 0.87) \\ 4(-1.45) + (10\times 0.87) \end{bmatrix} = \begin{bmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{bmatrix}$$

$$\boxed{mx = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}}$$

$$\hat{y} = mx + b = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

* Calculate error $\hat{y} - y$

$$\hat{y} - y = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

* MSE: $J = \frac{1}{2}\left((1-1.7851)^2 + (1.3447)^2\right)$

$$J = \frac{13.0066}{2}$$

$$\boxed{J = 6.5033}$$

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② Calculate numerical gradients

* gradient for m

$$\frac{\partial J}{\partial m} = \frac{2}{N} X^T (\hat{y} - y), \quad X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

Since $N = 2$

$$\frac{\partial J}{\partial m} = \frac{2}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times (-2.85) + (4 \times (-2.21)) \\ 3(-2.85) + (10\ (-2.21)) \end{bmatrix}$$

$$= \begin{bmatrix} -2.85 - 8.84 \\ -8.85 - 22.10 \end{bmatrix}$$

$$\boxed{\frac{\partial J}{\partial m} = \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}}$$

* gradient for b

$$\frac{\partial J}{\partial b} = \frac{2}{N} (\hat{y} - y)$$

$$\frac{\partial J}{\partial b} = \frac{2}{2} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} \Rightarrow \boxed{\frac{\partial J}{\partial b} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}}$$

③ Update parameters m and b

New = old − (α × gradient) with $\alpha = 0.01$

* Update m

$$m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - \begin{bmatrix} -0.1169 \\ -0.3065 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 + 0.1169 \\ 0.87 + 0.3065 \end{bmatrix}$$

$$\boxed{m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}}$$

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* Update b.

$$b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - 0.01 \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - \begin{bmatrix} -0.0285 \\ -0.0221 \end{bmatrix}$$

$$= \begin{bmatrix} 0.99 + 0.0285 \\ 0.89 + 0.0221 \end{bmatrix}$$

$$\boxed{b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}}$$

End of iteration 2

$$\boxed{m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}}$$

$$\boxed{b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}}$$

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2nd iteration:

* features $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

* targets $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

* weights $m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}$

* bias $b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$

1. Compute predicted values

$\hat{y} = mx + b$

* calculate mx

$$mx = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} = \begin{bmatrix} 1\times(-1.3331) + (3\times 1.1765) \\ 4\times(-1.3331) + (10\times 1.1765) \end{bmatrix}$$

$$mx = \begin{bmatrix} -1.333 + 3.5295 \\ -5.3324 + 11.7650 \end{bmatrix}$$

$$mx = \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix} //$$

* $\hat{y} = mx + b = \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$

$$= \begin{bmatrix} 2.1964 + 1.0185 \\ 6.4326 + 0.9121 \end{bmatrix} \quad \hat{y} = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix}$$

* Calculate Error

$$\hat{y} - y = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\hat{y} - y = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} //$$

* MSE : $J = \frac{1}{2}\left((-1.7851)^2 + 1.3447^2\right) = \frac{4.99479}{2} = 2.4974$ //

2. Calculate numerical gradients

* gradient for m

$$\frac{\partial J}{\partial m} = \frac{2}{N} X^T (\hat{y} - y), \quad X^T = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}$$

$$\frac{\partial J}{\partial m} = \frac{2}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix} = \begin{pmatrix} 1 \times (-1.7851) + (4 \times 1.3447) \\ 3 \times (-1.7851) + (10 \times 1.3447) \end{pmatrix}$$

$$\frac{\partial J}{\partial m} = \begin{pmatrix} -1.7851 + 5.3788 \\ -5.3553 + 13.447 \end{pmatrix} = \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix}$$

gradient for b

$$\frac{\partial J}{\partial b} = \frac{2}{N} (\hat{y} - y) \Rightarrow \frac{2}{2} \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

$$\frac{\partial J}{\partial m} = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

3. update parameters m & b

new = old − (α * gradient) , $\alpha = 0.01$

* update m

$$m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - 0.01 \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix} = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - \begin{bmatrix} 0.0359 \\ 0.0809 \end{bmatrix}$$

$$m = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} //$$

* update b

$$b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - 0.01 \begin{bmatrix} -1.785 \\ 1.3447 \end{bmatrix} = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - \begin{bmatrix} -0.0179 \\ 0.0134 \end{bmatrix}$$

$$= \begin{bmatrix} 1.0185 + 0.0179 \\ 0.9121 - 0.0134 \end{bmatrix} \Rightarrow b = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix} //$$

Trend of m, b and the error

① m and b

The numbers for m do not just change in one straight ~~line~~ direction. At first, they take a huge jump. for example, m drops all the way from 2 to 0.87. After that, it bounces back up to 1.17 and moves a little bit to 1.09. This back-and-forth movement happens because the first step was too big and stepped past the perfect answer. Now the maths is taking smaller steps back and forth to find the exact right spot. The bias numbers (b) make much smaller changes, meaning they were already closer to the right answer.

② Yes, the numbers are moving in a direction that lowers the errors. Our math proves this: the starting error was very high at 61.0. After the first update, it dropped to 6.5. By the third update, the error fell again to 2.5. This shows that the formula is working perfectly to fix the numbers and make the predictions much more accurate.

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